

Task 1 — Five Common Vulnerabilities

No.	Vulnerability	Description	Possible Impact
1	Broken Access Control	A user may be able to access another user's data or functionality without proper authorization checks.	Unauthorized access to accounts, orders, profiles, or other sensitive information.
2	SQL Injection	Unsafely handled database input can allow unintended SQL commands to be processed by the application.	Data disclosure, modification, or potentially unauthorized database access.
3	Cross-Site Scripting (XSS)	An application improperly handles untrusted content that is displayed in a user's browser.	Malicious scripts could execute in another user's browser and potentially steal session information or manipulate pages.
4	Authentication Weaknesses	Weak password policies, poor session handling, or flawed authentication mechanisms can make accounts easier to compromise.	Account takeover, unauthorized transactions, or exposure of personal information.
5	Security Misconfiguration	Systems may contain unnecessary services, default settings, excessive permissions, or improperly configured security controls.	Increased attack surface, information disclosure, or unauthorized access.

Task 2 — Install Nessus Essentials

Step 1 — Download Nessus

Download Nessus from the official Tenable website and select the appropriate installer for your operating system.

After installation, Nessus normally provides a local web interface.

Open a browser and go to:

- <https://localhost:8834>

You may see a certificate warning because Nessus uses HTTPS locally. Continue to the Nessus interface.

Step 2 — Create your account

During the setup:

1. Select Nessus Essentials.
2. Enter the required registration information.
3. Obtain the activation/registration code through Tenable's official process.
4. Create your Nessus administrator account.
5. Allow Nessus to initialize/download its plugins.

The initial plugin initialization can take some time.

Step 3 — Log in

- After initialization, log in using the account you created.

Task 3 — Scan Your Own Laptop or VMware VM

Step 1 — Find the target IP

On the Windows test VM:

- `ipconfig`

Look for:

- IPv4 Address

For example:

- IPv4 Address : 192.168.29.100\

Step 2 — Check connectivity

From the machine running Nessus:

- ping <TARGET-IP>

For example:

- ping 192.168.29.100

Step 3 — Create a Basic Network Scan

Open Nessus:

- <https://localhost:8834>

Then:

- New Scan → Basic Network Scan

Configure it approximately as follows:

- Name:
- Windows VM Vulnerability Scan
- Targets:
- <TARGET-IP>

Step 4 — Save and launch

Click:

- Save → Launch

Nessus will begin examining the authorized target.

Depending on the VM and scan configuration, the scan can take some time.

You can monitor its progress from the Scans page.

Step 5 — Examine the Results

When the scan finishes, open the scan.

You'll typically see vulnerability information organized by severity, such as:

- Critical
- High
- Medium
- Low

Click individual findings to examine:

- Vulnerability name
- Severity
- Affected service/software
- Description
- Evidence
- Recommended remediation

Step 6 — Export the Report as PDF

From the completed scan, use the available Report/Export option and select:

- PDF

Save it with a clear filename such as:

- Nessus_Vulnerability_Scan_Report.pdf

Submit

Your submission should contain:

- Nessus dashboard screenshot
- Target IP
- Screenshot of completed scan
- Vulnerability findings
- Generated PDF report

Task 4 — Nessus vs OpenVAS

Feature	Nessus	OpenVAS / Greenbone
Scanner & management	Nessus provides an integrated vulnerability-scanning platform with a web interface.	OpenVAS is part of the Greenbone vulnerability-management ecosystem.
Vulnerability coverage	Uses Tenable's vulnerability/plugin ecosystem and frequent updates.	Uses the Greenbone Community Feed for vulnerability testing content in the community ecosystem.
Licensing	Nessus has different commercial offerings, with Nessus Essentials available for limited free use.	Greenbone provides community/open-source components, while commercial Greenbone products provide additional capabilities and support.